

MARS

Malware Analysis & Reverse-engineering System

ANALYSIS ID	REDESIGN_001
TARGET FILE	malware_sample.exe
GENERATED	2026-06-19

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1. File Intake Summary

Metadata captured at ingestion time. Hash values are computed immediately on submission before any analysis step touches the file.

Analysis ID	REDESIGN_001 <i>Unique identifier auto-generated for this analysis session.</i>
Filename	malware_sample.exe <i>Original filename of the submitted sample as read from disk.</i>
Extension	.exe <i>File extension used to determine the analysis pipeline branch.</i>
File Size	40960 bytes <i>Total size of the submitted file in bytes.</i>
SHA256	aabbccdde112233445566778899aabbccdde112233445566778899aabb1122 <i>SHA-256 cryptographic hash — primary unique identifier for the sample.</i>
Submission Timestamp	2026-06-19 12:00:00 <i>UTC date and time at which the file was submitted to the pipeline.</i>

2. Package & Archive Unpacking

Extracted 2 file(s) — 1 **flagged** as executable payloads.

Total Extracted Files	2 <i>Number of files unpacked from the archive or MSI installer payload.</i>
Flagged Payloads	1 <i>Count of extracted files whose extension matches a high-risk executable type (.exe, .dll, .sys).</i>

Extracted Artefacts

Path	Ext	Size (B)	SHA-256 (abbrev.)	Status
dropper/payload.exe	.exe	8192	deadbeefdeadbeef...	FLAGGED
config/settings.ini	.ini	512	cafebabecafebabe...	clean

3. Deep Static Analysis

PE file inspection performed without executing the sample. Covers header metadata, security mitigations, section entropy, IAT analysis, manifest parsing, string extraction, and YARA signature matching.

Target: malware_sample.exe

PE Headers

Machine Architecture	0x14c <i>Target CPU architecture encoded in the COFF File Header (e.g. 0x14c = x86, 0x8664 = x64).</i>
Compile Timestamp	2025-01-15 08:30:00 <i>Date/time the binary was linked. Stored in the PE File Header — trivially forgeable.</i>
Address of Entry Point	0x1234 <i>Relative virtual address (RVA) of the first instruction executed. Unusual RVAs suggest packing.</i>

Image Base	0x400000 <i>Preferred virtual address where the image is loaded. 0x400000 is the traditional default.</i>
Number of Sections	5 <i>Count of PE sections. Values outside 3-10 are often associated with packers or malformed binaries.</i>

Mitigations

DEP / NX Bit	Enabled <i>Data Execution Prevention — marks data pages non-executable to block shellcode.</i>
ASLR	Disabled <i>Address Space Layout Randomisation — randomises load addresses to hinder ROP chains.</i>
Stack Canaries (/GS)	Disabled [WARNING] <i>Compiler-inserted stack cookie; detects stack-buffer overflows before return.</i>

Sections

Section: .text	Perms: R-E Entropy: 6.12
Section: .data	Perms: RW- Entropy: 7.85 [PACKED/HIGH ENTROPY]

Suspicious Imports

Tracked APIs Found	3 <i>Number of monitored Win32 API imports found in the Import Address Table (IAT).</i>
APIs	CreateRemoteThread, VirtualAllocEx <i>Comma-separated list of suspicious API names. Cross-reference with MITRE ATT&CK; technique IDs.</i>

YARA Signatures

Hits	2 <i>Total number of YARA rule signatures that matched against the sample.</i>
Matched Rules	Mitre_T1055_ProcessInjection <i>YARA rule names that fired. Each name encodes a threat family or behavioural pattern.</i>

4. Dynamic Sandbox Analysis

Behavioural telemetry captured during live detonation in the VMware sandbox. The in-guest agent instruments filesystem, registry, network, process, memory, and hardware subsystems in real time.

Target: malware_sample.exe

Telemetry Summary

Category	Events	Notable / High-Risk
Filesystem Activity (FR-DYN-01)	1	1 flagged
Registry Mutations (FR-DYN-02)	1	?
Persistence Mechanisms (FR-DYN-03)	1	1 flagged
Process Activity (FR-DYN-04)	1	1 flagged

Category	Events	Notable / High-Risk
Memory Forensics (FR-DYN-05)	0	?
Network Telemetry (FR-DYN-06)	2	2 flagged
Hardware / System Stress(FR-DYN-07)	0	?
System & Agent Events	1	?

Event Detail

Filesystem Activity (FR-DYN-01) (1 event)

File creation, deletion, and modification events observed during live sandbox detonation.

- **FILE_DROP: C:/Windows/Temp/svchost32.exe**

Registry Mutations (FR-DYN-02) (1 event)

Windows Registry key reads, writes, and deletions captured during execution.

- REG_SET: HKCU/Run/updater

Persistence Mechanisms (FR-DYN-03) (1 event)

Mechanisms used to survive reboots: Run keys, scheduled tasks, service registration.

- **REG_RUN_KEY: HKCU Run -> svchost32.exe**

Process Activity (FR-DYN-04) (1 event)

Child processes spawned and process-injection events observed by the in-guest agent.

- **PROCESS_INJECT: lsass.exe (PID 840)**

Network Telemetry (FR-DYN-06) (2 events)

DNS queries, HTTP/S requests, and raw TCP/UDP connections intercepted by the sandbox NIC.

- **DNS: evil-c2.example.com**
- **HTTP: GET /payload.bin**

System & Agent Events (1 event)

OS-level events including privilege escalation, UAC bypass attempts, and driver loads.

- UAC bypass attempted